Wall follower using Proportional Derivative Control

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What is Wall follower?

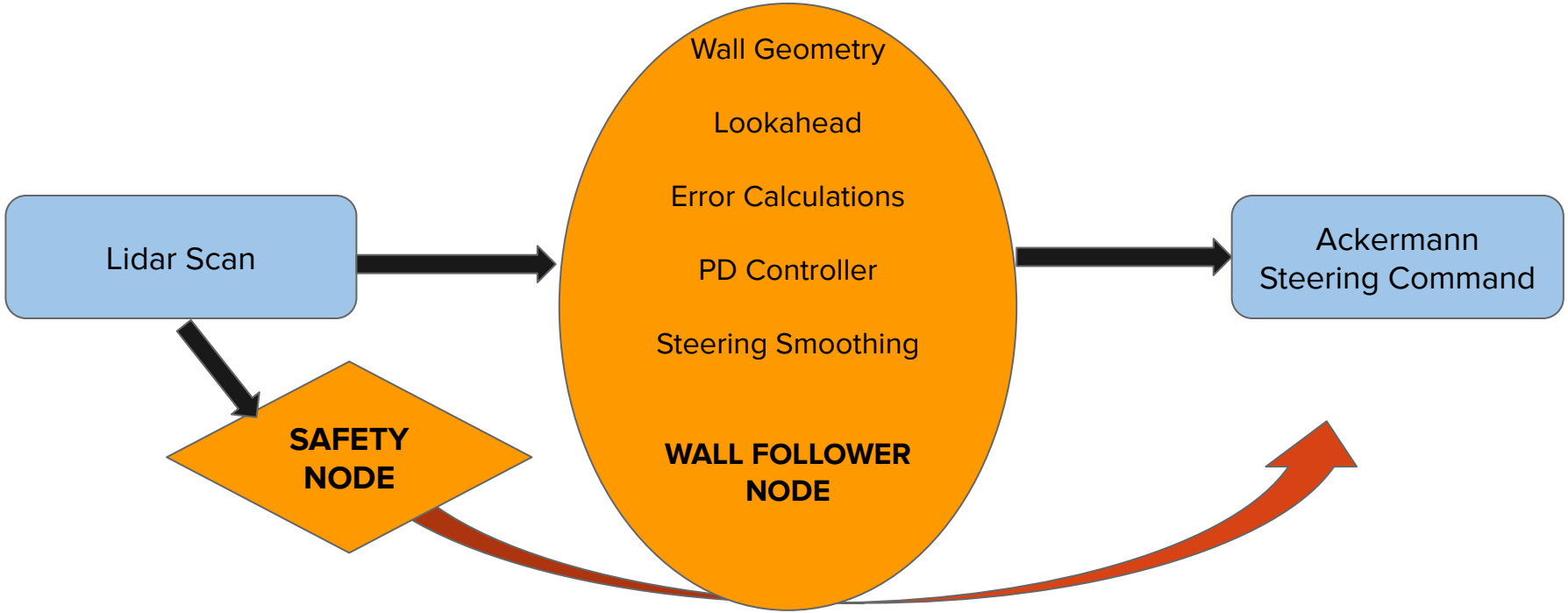
Wall Follower Performance Objectives

Our goal was to evaluate **how well** the wall follower maintained a consistent distance from the wall while driving safely and smoothly.

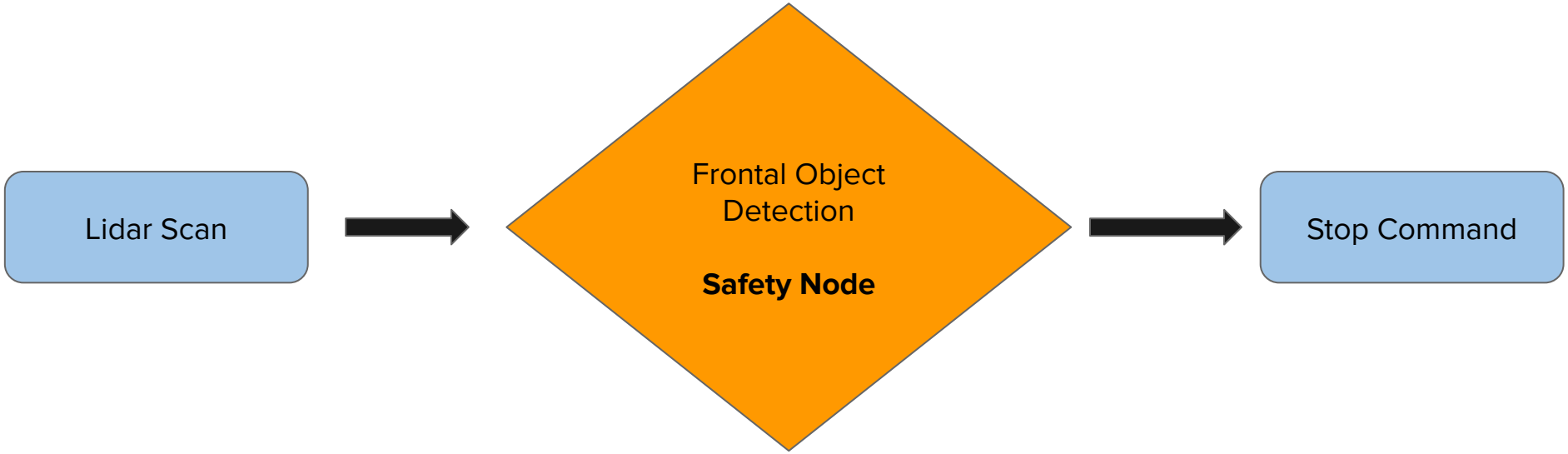
Metrics we focused on:

- **Pass our created test cases**
- **Mean Error under 0.5 m**
- **Standard Deviation of Error**
- **Safe Operation (Braking Distance)**

Wall follower PD Implementation



Safety Node Implementation

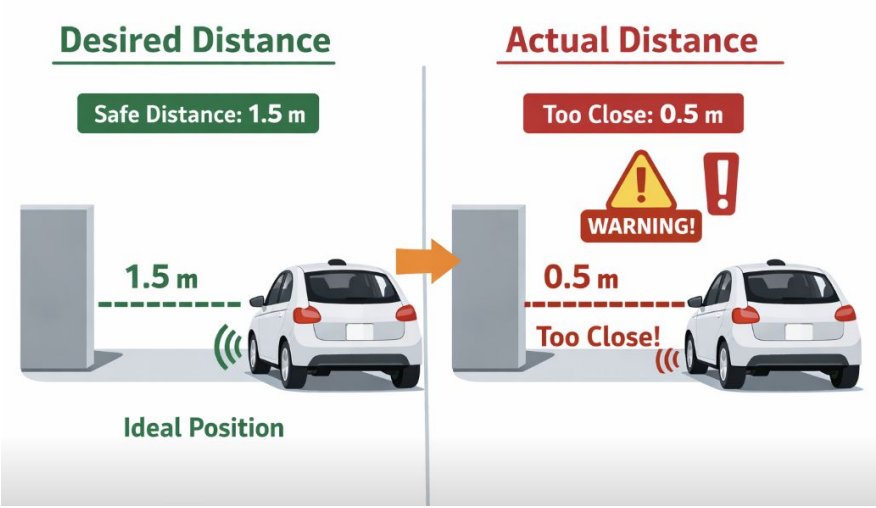


What is error in the experiment???

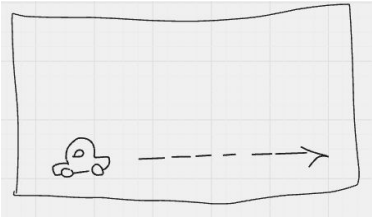
Error = Desired distance − Actual distance

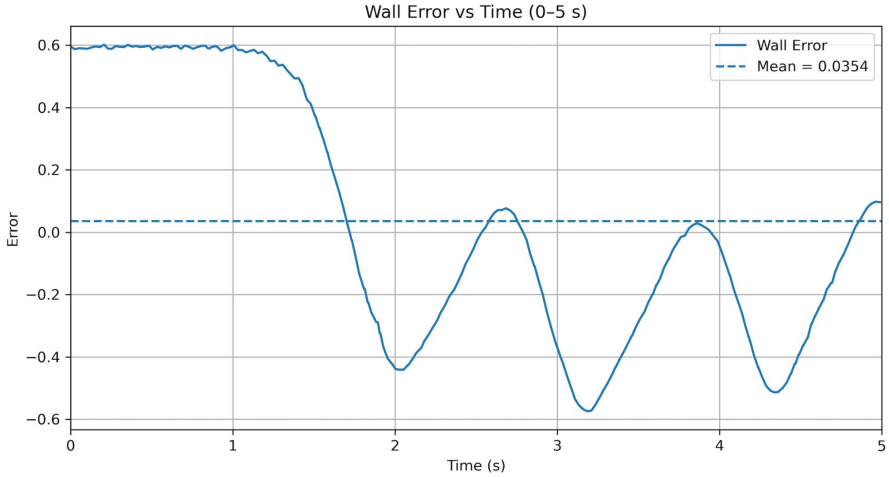
Error Interpretation

Error > 0 ➔ car too close to wall
Error < 0 ➔ car too far from wall



Straight Line – Speed 1m/s ✅

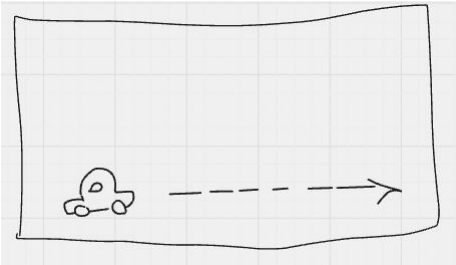


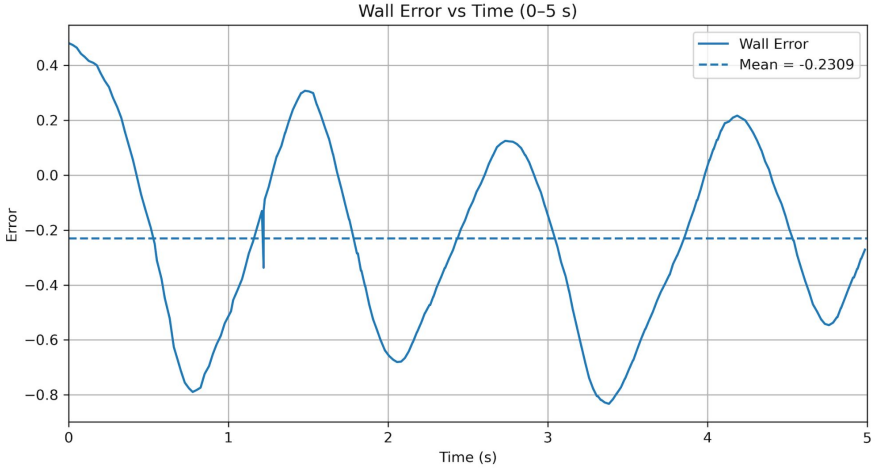


Mean Error : 0.035 m

Standard Deviation : 0.18 m

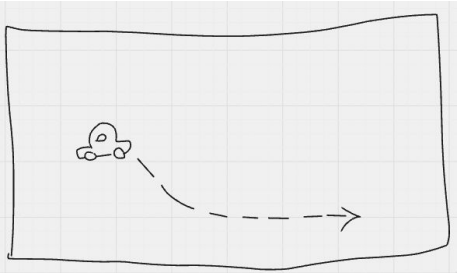
Straight Line – Speed 3m/s ✅

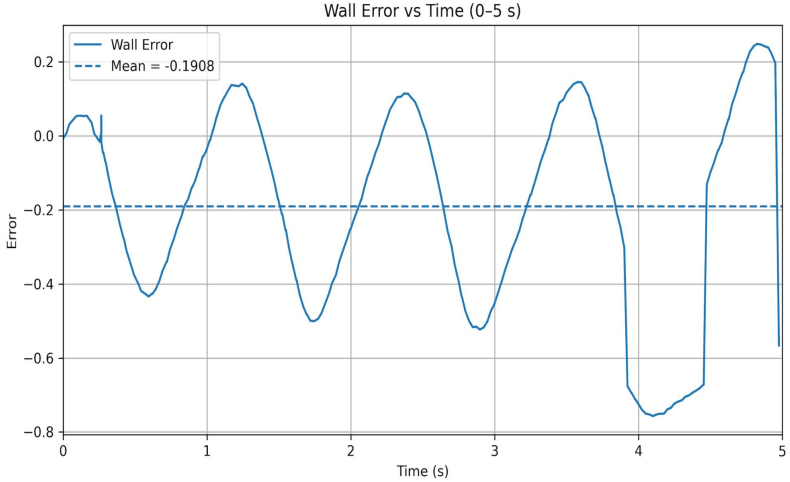




Mean Error : -0.23 m
Standard Deviation : 0.33 m

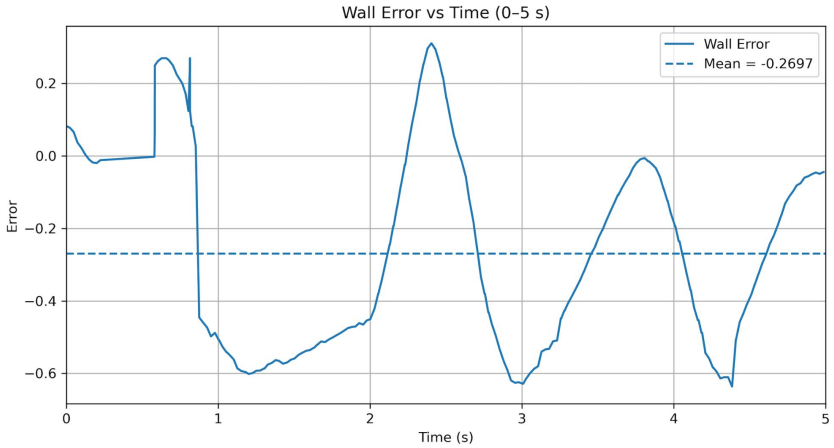
Start in Middle - Speed 1m/s ✅





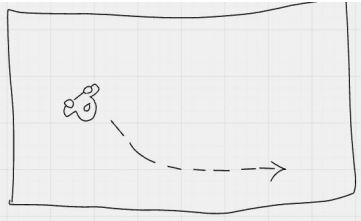
Mean Error : -0.19 m
Standard Deviation : 0.277 m

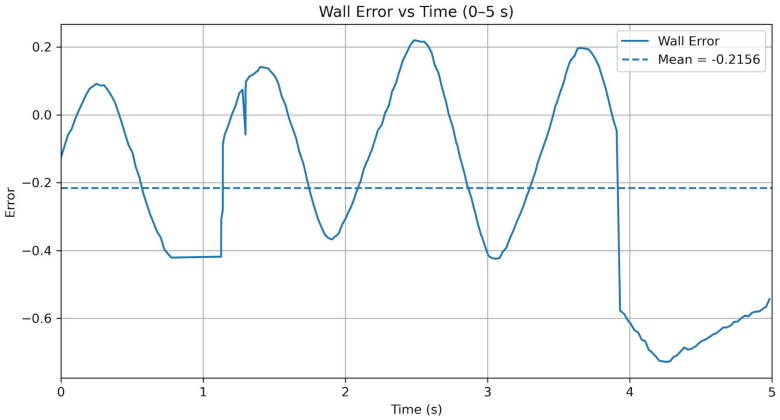
Outer corner - Speed 1m/s ✅



Mean Error : -0.26 m
Standard Deviation : 0.28 m

Facing away from wall - Speed 1m/s ✅

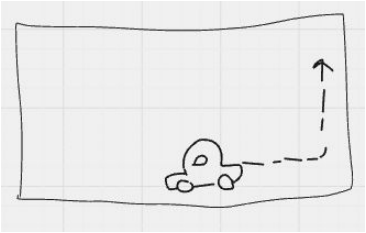




Mean Error : -0.22 m

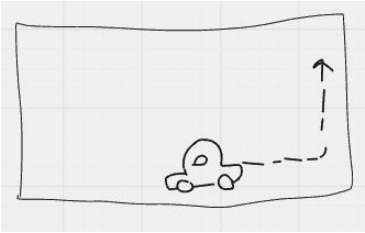
Standard Deviation : 0.29 m

Inner corner

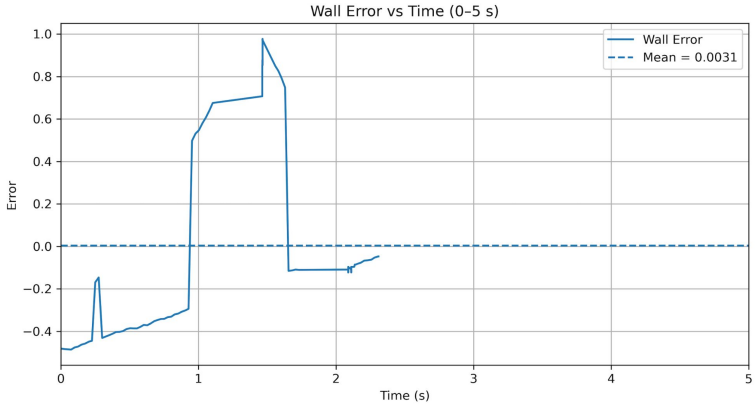


Controller Unable to Maintain wall following at Inner corner

Inner corner - Speed 1m/s 👀



Controller Unable to Maintain wall following at Inner corner

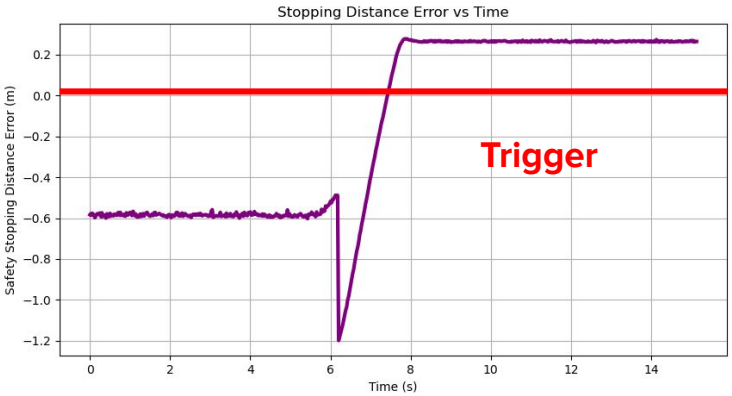


Mean Error : 0.003 m

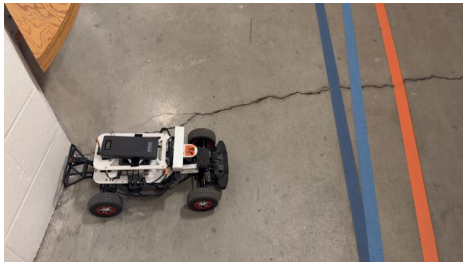
Standard Deviation : 0.45 m

Stop!!!

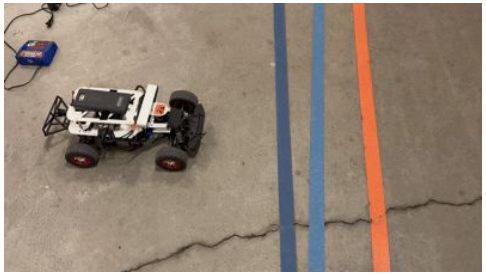
Speed: 1 m/s



Minimal Distance Threshold: 0.5 m



Minimal Distance Threshold: 1.3 m



Conclusions & Future Work

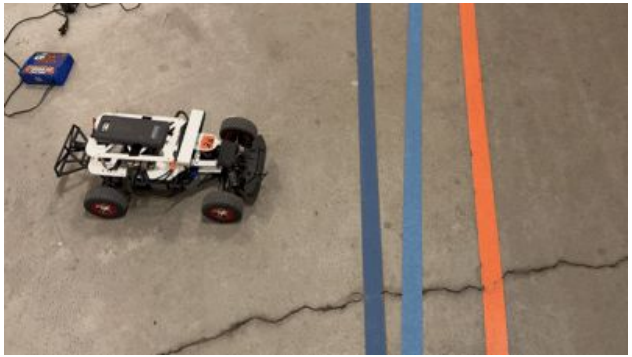
Achievements

- Successful wall-following using LiDAR and PD control
- Stable tracking at low speed

Lessons Learned

- Controller tuning strongly affects stability
- Higher speeds introduce oscillations

Future Work

- Wall-loss recovery strategy to pass test like inner corner turning
- Speed-dependent control tuning

Time for the Questions!